MIL-C-915/8D
1 August 1972
SUPERSEDING
MIL-C-00915/8C(SHIPS)
20 March 1972
MIL-C-915/8B
24 February 1969

MILITARY SPECIFICATION SHEET

(D) CABLE, ELECTRICAL, 600 VOLTS, TYPES DSS, TSS, FSS AND 7SS

(D) This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

Construction (Watertight)

First - Copper conductor, tin coated. See Details for size.
Second - Synthetic rubber insulation. See Details for thickness. Special identification code, applied by Method 3.
Third - The required number of conductors (see Details) cabled together with a lay length not greater than that specified in the Details. The insulated conductors shall not adhere to one another. Rubber fillers as necessary in order to form a firm, round assembly. Fillers in outer valleys may be extruded in order to be integral with the belt (see Fourth).
Fourth - Belt of synthetic rubber, nominal wall thickness 0.030-inch, over the assembled conductors. Belt and fillers shall be capable of removal from the assembly without damaging the insulation on conductors.
Fifth - Braided shield of AWG No. 34 tin coated copper. Braid angle of 50 degrees, minimum; coverage of 90 percent, minimum.
Sixth - Jacket of either polychloroprene or chlorosulfonated polyethylene, black. See Details for thickness. If necessary in order to meet the electrical requirements for the completed cable, the jacket may consist of two layers bonded together, with the inner layer consisting of synthetic rubber insulation. Cable surface marking required.

NOTE: Manufacturer's identification tape may be omitted, at the manufacturer's option, from all types and sizes.

Details

Type and Size	Conductor size, Navy Standard	Number of conductors	Insulation[1/] thickness minimum (inch)	Lay length (inch)	Jacket[1/] thickness minimum (inch)	Overall diameter Min. (inch)	Overall diameter Max. (inch)	Conductor resistance per 1000 ft. max., (ohms)
DSS-2	2(7)	2	0.020	2.0	0.050	0.370	0.390	6.64
DSS-3	3(7)	2	.025	2.0	.080	.480	.500	4.15
DSS-4	4(7)	2	.025	2.5	.065	.480	.500	2.57
TSS-2	2(7)	3	.020	2.0	.050	.385	.400	6.64
TSS-3	3(7)	3	.025	2.5	.070	.480	.500	4.15
TSS-4	4(7)	3	.025	2.5	.055	.480	.500	2.57
FSS-2	2(7)	4	.020	2.5	.080	.480	.500	6.64
FSS-3	3(7)	4	.025	2.5	.060	.480	.500	4.15
FSS-4	4(7)	4	.025	3.0	.085	.600	.625	2.57
7SS-2	2(7)	7	.020	3.0	.085	.600	.625	6.64

1/ Because of the compacted construction necessary in order for these cables to meet the required hydrostatic tests, centering and circularity requirements for insulation and jacket are waived, provided that minimum wall thicknesses are within the specified limits, and that the minimum thickness at any cross section is not less than 80 percent of the specified minimum thickness.

(D) Denotes changes

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:05Z
Check the source to verify that this is the current version before use.

EXAMINATION AND TESTS:

	Requirements:
Basic electrical:	
Voltage withstand - Volts, rms, minimum	
Conductor to conductor (5 minutes)	3000
Conductor to shield (5 minutes)	1000
Shield to water (5 minutes)	500
Insulation resistance - Megohms/1000 ft., minimum	
Conductor to conductor	500
Conductor to shield	500
Shield to water	250
Conductor resistance - Ohms/1000 ft. at 25°C., maximum	(See Details)
Group A:	
Visual and dimensional	
Hydrostatic (open end) - At 500 psi, in 2 hours, cu. in., max.	0
Mutual capacitance (type DSS only) - pF/ft. at 1 kHz, max (measurement to be made with black conductor connected to shield at one end of specimen)	45
Group B:	
(D) Bending endurance - At minus 40° ± 2°C. on 4-inch maximum diameter mandrel, cycles, minimum	1000
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - psi, minimum	600
Elongation - Percent, minimum	350
Jacket (cable)	
Tensile strength - psi, minimum	1800
Elongation - Percent, minimum	300
Set - Inch, maximum	3/8
Group C	
Physicals (aged)	
Insulation (conductor)	
Air pressure heat	
(D) Tensile strength - Percent of unaged, minimum	65
(D) Elongation - Percent of unaged, minimum	65
Jacket (cable)	
Hot oil immersion	
(D) Tensile strength - Percent of unaged, minimum	65
(D) Elongation - Percent of unaged, minimum	65
Permanence of printing (jacket) - Cycles, minimum	250

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical plus all of Groups A, B and C plus the following:

Hydrostatic (open end), after bending endurance - At 500 psi, in 2 hours, cu. in. maximum	0
Pressure cycling (DSS-3 only) - At 500 psi, percent of variation in mutual capacitance, maximum	10
Insulation resistance (after 90-day immersion in tap water) - Megohms/1000 ft., minimum	
Conductor to conductor	500
Conductor to shield	500
Shield to water	250

UNIT ORDERING LENGTHS:
All types and sizes: 500 ft. (nominal) coils.

(D) Custodians:
Army - MI
Navy - SH
Air Force - 80

(D) Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

(D) User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0609-67)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T12:05Z
Check the source to verify that this is the current version before use.